

YONG KHENG BEH

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Personal Website: kheng2023.github.io | GitHub: github.com/Kheng2023

TECHNICAL SKILLS

- Programming Languages: Python, C++, C, SQL, HTML, CSS, JavaScript
- Data Analysis & AI: Pandas, Matplotlib, Scikit-learn, LangChain
- Databases: MySQL, OpenSearch, hnswlib, FAISS, ScaNN, Usearch
- Tools & Platforms: Linux, Docker, Github

WORK EXPERIENCE

YOURANSWER INTERNATIONAL PTY LTD

Jul 2024 – Oct 2024

Software Development Intern

Key Skills: Python, Bash (Linux), AI/ML (Embeddings, Vector Search), Docker, OpenSearch

- Developed an automation tool in Python to evaluate semantic search performance, streamlining future research and testing.
- Experimented with top-ranked embedding models from the MTEB leaderboard, as well as different chunking methods (e.g., sentence, recursive, token) and vector stores (e.g., FAISS, HNSW) to determine the optimal combination for search performance (accuracy, speed, memory use).
- Identified and fixed a logic flaw in the company's core search algorithm causing backward rankings, resulting in a 10% improvement in semantic search accuracy.
- Presented experimentation results to the research team, effectively communicating key findings and recommendations for future development.
- Authored comprehensive technical documentation, ensuring knowledge transfer.

CONNECTING UP, INFOXCHANGE

Mar 2024 – Jun 2024

Industry Project Intern

Key Skills: Python, LangChain, Flask, Docker, Agile Software Development

Links: [My Internship Experience](#), [Chatbot Project Report](#), [GitHub Repo](#)

- Led backend development for an AI-powered chatbot, improving search functionality for a 15,000+ entry database at a not-for-profit organization.
- Integrated LangChain for Retrieval-Augmented Generation (RAG), utilising Large Language Model to enhance chatbot query understanding and response accuracy by at least 15%.
- Built the backend with Flask & Flask-SocketIO, integrating WebSocket for real-time communication.
- Deployed the chatbot using Docker, mitigating compatibility issues in legacy software environment.
- Conducted rigorous testing, debugging, and performance optimizations, resulting in a reduction in error rates.

AUSTRALIAN INSTITUTE OF MACHINE LEARNING (AIML)

Dec 2023 – Feb 2024

Summer Research Project Scholarship Intern

Key Skills: Python, Data Analysis (Pandas, Matplotlib, Seaborn), Machine Learning (Scikit-learn)

- Collaborated with healthcare experts to explore AI-driven decision-making solutions for medical applications.
- Processed and analysed a large-scale healthcare dataset with 6,000+ patient records, training machine learning models to predict patient outcomes.

- Conducted statistical analysis & data preprocessing to improve model accuracy.
- Researched and reviewed AI-driven predictive analytics in healthcare, integrating insights into model design.

UNIVERSITY OF ADELAIDE

Jul 2023 – Oct 2023

Peer Assisted Study Sessions (PASS) Leader

Key Skills: Communication, Teaching, Leadership, Collaboration, Python

- Mentored approximately 5 students per week for 12 weeks in Python programming, providing guidance and support to enhance their learning experience and help them succeed in their coursework.
- Consistently received positive feedback from students, with an average teaching quality rating of 6.5 out of 7 in end-of-course reviews.

EDUCATION

UNIVERSITY OF ADELAIDE

Feb 2023 – Dec 2024

Master of Computing and Innovation

[GPA: 6.83/7.00]

Relevant Courses: Operating Systems (C, C++), Algorithms and Data Structures (C++), Computer Systems, Computer Networks & Applications (C, C++), Introduction to Statistical Machine Learning (Python)

UNIVERSITY PROJECT

Rail Break Prediction Model Development | Insightfactory.ai

Jul 2024 – Oct 2024

Key Skills: SQL (PySpark), Data Analysis (Pandas, Matplotlib, Seaborn), Machine Learning (Scikit-learn, XGBoost)

- Developed a machine learning model to predict the likelihood of rail breaks 30 days in advance, sponsored by InsightFactory.ai.
- Processed and analysed large-scale ARTC railway data using SQL (PySpark), including data cleaning, rebalancing, and preprocessing.
- Through experimentation with various machine learning models (including linear regression, random forest, and XGBoost), identified the optimal approach for forecasting breaks, achieving a 52% F1 score (5th in the leaderboard out of 15, a top performance) in predicting rail breaks 30 days in advance.

VOLUNTEER EXPERIENCE

CONNECTING UP, INFOXCHANGE

Jan 2025 – Present

Website Redevelopment Volunteer

Link: [My Volunteer Experience](#)

Key Skills: Drupal, mySQL, Python, LangChain

- Collaborating with a team of volunteers to redevelop the SAcommunity.org website, improving functionality and user experience.
- Exploring solutions to integrate the AI-powered chatbot into the new website for enhanced accessibility.

PERSONAL PROJECT

PERSONAL WEBSITE

Jan 2025 – Present

Link: [kheng2023.github.io](#)

Key Skills: HTML, CSS, JavaScript

- Designed and built a personal website using HTML and CSS.
- Currently learning JavaScript to enhance website functionality with interactive elements and dynamic content.